

[!] PROJECT DOOMSDAY

MULTI-AGENT BLACK SWAN STRESS-TESTING ENGINE

"Every portfolio has a breaking point. We find it in 150 seconds."

5

AI AGENTS

6

PARALLEL SEARCH VECTORS

<60s

ANALYSIS TIME PER AGENT

BUILT WITH: GOOGLE GEMINI 2.0 FLASH |
TAVILY | STREAMLIT | PYTHON

MOOSA TALHA AL KASERI

BLIND SPOTS KILL PORTFOLIOS

[01] NORMAL DISTRIBUTION FAILURE

Traditional models assume gaussian returns – they collapse precisely when tail risk materialises. 2008, COVID, SVB were not outliers. They were the model's blind spot.

[02] SILOED RISK IDENTIFICATION

Analysts identify risks in isolation, missing second-order cascades and geographic convergence. Risk [A] triggering Risk [B] triggering Risk [C] is invisible to current tools.

[03] THE TOOLING GAP

Existing tools are either too simplistic (flat % haircuts), too slow (weeks of manual scenario work), or black boxes with zero methodology transparency.

THE GAP

No tool exists that identifies specific, current Black Swan risks – validates them adversarially – maps geographic propagation – models contagion cascades – and computes distressed valuations. Transparently. In under 60 seconds.

93%

OF INSTITUTIONAL INVESTORS CITE TAIL RISK AS
#1 CONCERN
YET ONLY 12% HAVE SYSTEMATIC TOOLS TO MODEL
IT

– CFA INSTITUTE 2024 SURVEY

HOW IT WORKS: 7-STEP PIPELINE

PHASE 1: INTELLIGENCE GATHERING

01

USER INPUT

Ticker + Chaos Level (0-1)

02

WORLD STATE

Live VIX / Oil / Gold / Yields → Global Fear Class

03

COMPANY PROFILE

Smart ticker resolve · Auto USD normalisation

04

INTELLIGENCE SCAN

6 parallel Tavily vectors → Gemini synthesises 6 material risks

PHASE 2: ANALYSIS & OUTPUT

05

ADVERSARIAL TRIBUNAL

Bear ↔ Bull ↔ Judge — 3-agent debate per risk

06

GEOGRAPHIC MAPPING

Fracture nodes + curved convergence lines on Plotly globe

07

VALUATION + CASCADE

5-path methodology · Contagion propagation · Transparent waterfall



5 AI AGENTS IN COORDINATION · GEMINI 2.0 FLASH VIA GOOGLE AI STUDIO · FAILOVER: NVIDIA NIM + FIREWORKS AI

5 SPECIALIZED AGENTS IN COORDINATED SWARM

INTELLIGENCE ANALYST

RISK DISCOVERY

T=0.5

Synthesises 6 Tavily vectors + company fundamentals into geolocated risk scenarios with severity, probability & revenue-at-risk estimates.

BEAR ADVOCATE

PROSECUTION

T=0.6

Argues worst-case with historical precedents and catastrophic framing. Pushes severity UP. Adversarial counterweight.

BULL ADVOCATE

DEFENCE

T=0.6

Challenges evidence, presents mitigating factors, argues market has priced the risk in. Pushes severity DOWN.

FRACTURE JUDGE

FINAL VERDICT

T=0.3

Low-temperature, high-conviction. Calibrated rubric: 8+ catastrophic, 6-7 material, <4 dismissed. Final authority.

CONTAGION MODELER

CASCADE

T=0.5

Models 2nd / 3rd / 4th-order propagation. Traces how a shock cascades through P&L → credit → refinancing → capex.

- Single-agent = confirmation bias · adversarial architecture forces steel-manning before judgment
- Mimics real IC process: Research Analyst → IC Debate → PM Decision · Judge T=0.3 = calibrated low-variance verdicts

WHAT MAKES THIS DIFFERENT

D1

CONTEXT-AWARE VALUATION

5 distinct methodologies routed by company type. Banks get P/BV. Cyclical get normalised EBITDA. High-growth gets EV/Revenue. Market-anchored, stressed downward only.

D2

GEOGRAPHIC INTELLIGENCE

Every risk mapped to physical coordinates. Curved convergence lines show vulnerability concentration. Visual answer to: where on Earth is my exposure?

D3

CONTAGION CASCADES

Models Risk A $\rightarrow$ Effect B $\rightarrow$ Effect C propagation. Compound correlation ($\rho \rightarrow 1$) as chaos increases. Inspired by Kiyotaki-Moore credit cycles and Minsky Moments.

D4

FULL TRANSPARENCY

Every formula shown with numbers plugged in. Every routing decision flagged. Every agent reasoning visible. Same inputs $\rightarrow$ same methodology $\rightarrow$ same output.

DOOMSDAY vs. MARKET

FEATURE	OURS	PEERS
Adversarial validation	✓	✗
Real-time intelligence	✓	✗
Contagion cascade model	✓	✗
Methodology transparency	✓	✗
Indian market (NSE/BSE)	✓	✗
Sub-60s analysis	✓	Partial
Geographic risk map	✓	Partial
5 valuation engines	✓	1-2

FROM HACKATHON TO PRODUCT

\$65B

RISK ANALYTICS MARKET 2027

\$25K

BLOOMBERG SEAT / YR

\$5M

ANNUAL BESPOKE SCENARIO SPEND

FREE

\$0

Retail / Students

3 analyses/day · Lead generation

PRO

\$99-299/mo

Analysts / RIAs / Funds

Unlimited · PDF export · API access

INSTITUTIONAL

\$2K-10K/mo

Hedge Funds / PE / Family
Offices

Custom domains · SEBI filings · White-label

API-AS-A-SERVICE

Per-call

Fintech / Robo-advisors

Embed stress-test into existing products

UNIT ECONOMICS — PRO TIER

GEMINI API COST / ANALYSIS \$0.03-0.08

TAVILY COST / ANALYSIS \$0.01-0.02

GROSS MARGIN AT SCALE >95%

INDIAN MARKET (NSE/BSE) SUPPORT — MASSIVELY UNDERSERVED BY WESTERN
TOOLS — KEY MOAT

BUILT WITH — NEXT 3-6 MONTHS

PRIMARY AI

Gemini 2.0 Flash

Best latency/cost for structured JSON · Google AI Studio

FAILOVER AI

NVIDIA NIM (Llama 3.3 70B)

Auto-switch if rate-limited — zero user disruption

REAL-TIME SEARCH

Tavily API

Grounds AI in live events — no stale hallucinations

FINANCIAL DATA

yfinance

Live prices, fundamentals, FX — 60+ global exchanges

FRONTEND

Streamlit

Single Python file — deploy in minutes

VISUALISATION

Plotly

Interactive maps (Scattermapbox) · Waterfall charts

[ROADMAP — NEXT 3-6 MONTHS]

NEXT

SEC EDGAR 10-K Item 1A auto-parsing

NEXT

SEBI / BSE filing integration for Indian equities

Q3

Portfolio-level analysis + cross-holding correlation

Q3

PDF export: McKinsey-style IC memo

Q4

"Lying Gap" detection: news vs filing divergence = alpha signal

Q4

Macro feedback loops: Chaos >0.8 auto-adjusts VIX/Oil

Q4

Slack / Teams webhook alerts for watchlist stocks